Refuge Scenarios

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About This Document

The version of the IREF included in the grant proposal attempts to tackle some of Os Schmitz's early 2000's work on how prey should respond to predators with different predator hunting modes (e.g., sit-and-wait, sit-and-pursue, active hunting). However, it leaves out an important additional aspect of Os's work: the presence of refugia from predation. Thus, **an important goal of this document is to determine whether we can take the IREF, which is not a spatiotemporally explicit model, and incorporate refugia in a way that is analogous to Os's informal verbal models on the subject.**

Of particular importance to this question are the following figures, from his textbook (and Schmitz et al. 2004 Ecol Lett):

The IREF already covers the cases where the predator and prey fully overlap spatially, i.e., the left two quadrants, and arguably the lower right quadrant, although in a somewhat coarse way. This document will largely be focused on tackling the cases in the upper right quadrant, where the prey's spatial domain is larger than the predator's and therefore includes predator-free refugia. In these cases, Os argues that the prey should shift its habitat usage to the refuge regardless of the predator's hunting mode, and the system should be dominated by TMIIs except with sit-and-pursue predators, where it may also feature a blend of the two.

It should be noted as well that Os's version of a sit-and-wait predator may be somewhat rare in most terrestrial systems, because arguably a predator that does not move at all to locate prey is similar to a filter feeder. What we've been calling a "sit-and-wait" predator he might actually call a "sit-and-pursue" predator.

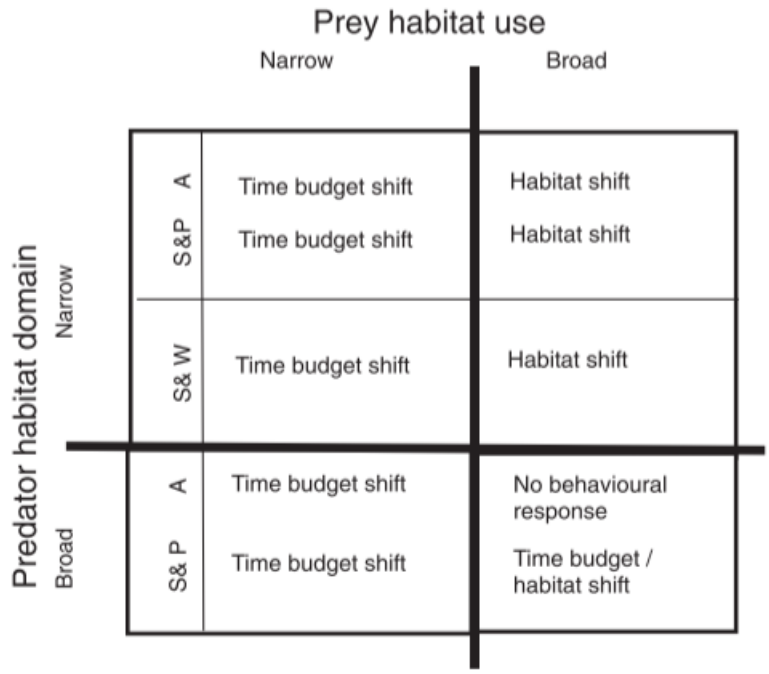


Figure 2 Synthesis of prey antipredator responses in relation to prey and predator habitat domain and predator hunting mode. S & P, A and S & W are sit-and-pursue, active and sit-and-wait hunting strategies, respectively. Species with a narrow habitat domain select only part of the entire available habitat. Species with a broad habitat domain use the entire range of habitat.

Figure 1: Fig. 2 from Schmitz et al. 2004

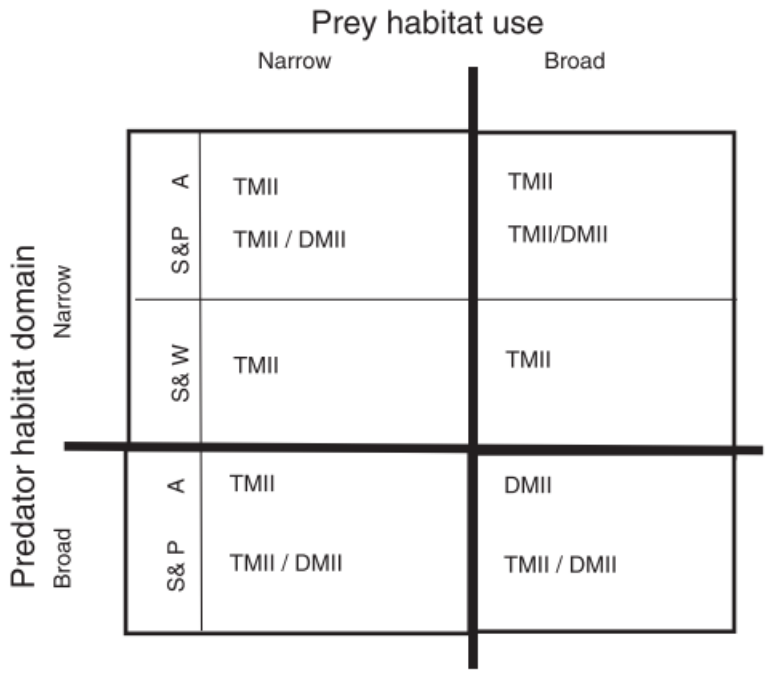


Figure 3 Hypothesized nature of emergent trophic cascade in cases involving different prey behavioural responses to predators with different hunting modes and habitat domains. S & P, A and S & W are sit-and-pursue, active and sit-and-wait hunting strategies, respectively. Species with a narrow habitat domain select only part of the entire available habitat. Species with a broad habitat domain use the entire range of habitat.

Figure 2: Fig. 3 from Schmitz et al. 2004

Model Description

For the following scenarios, we will explore a predator-prey system with a refuge from predation using the IREF.

In brief, we will use the IREF to calculate the optimal amount of time for prey to spend in the refuge habitat (i.e., the proportion of time spent in the refuge will be the trait on the x-axis). To do so, we will assume that prey have some constant body velocity inside the refuge, and a separate constant velocity outside the refuge. We will derive these constant values for velocity by applying the IREF to each of the two habitat types separately.

In both sets of scenarios, we will use the following equation for calculating encounter rate (Z) between the predator and prey:

$$Z = 2rN * sqrt(v^2 + s^2)$$

where s is prey velocity (i.e., displacement rate), v is predator velocity, N is predator density, and r is the radius within which the predator can perceive prey. We will use this as a proxy for the predator's average attack rate. This is similar to assuming that the predator has a Type I functional response, but in actuality, we are simply ignoring the effects of predator satiation on the interaction, because the IREF is not a temporally explicit model. A Type II functional response could be incorporated into the model, but it would add another parameter and would probably simply make the mortality-trait curve a more saturating relationship.

For simplicity, detection radius is kept constant such that $2r = 1$. We will also keep predator velocity (v) constant. In the baseline IREF simulation (Fig. 2 in the grant), $v = 0.2$, so we will use this value.

This leads to a mortality-trait curve that is slightly concave up with respect to prey body velocity, because at high prey displacement rate, encounters are driven largely by prey velocity, whereas at low prey displacement rate, encounters are driven largely by predator velocity.

```
## The mortality-trait curve This forumula is simplified by
## setting some of the values as constants r is assumed to
## be 0.5 (which cancels out the 2), while k, v, and s are
## manipulated values k is predator density v is predator
## velocity s is prey velocity

pred.curve <- function(N_pred, k, v, s) {
    k * sqrt(v^2 + s^2)
}
```

The growth-trait curve is determined by the rate at which the prey consumes its resource, R. This takes the form of a Type II functional response:

$$f(R) = \frac{sR}{1 + shR}$$

where s is prey displacement rate, h is handling time, and R is resource density. This results in a growth curve that is strongly concave down, because higher body velocity has diminishing returns due to time spent handling food.

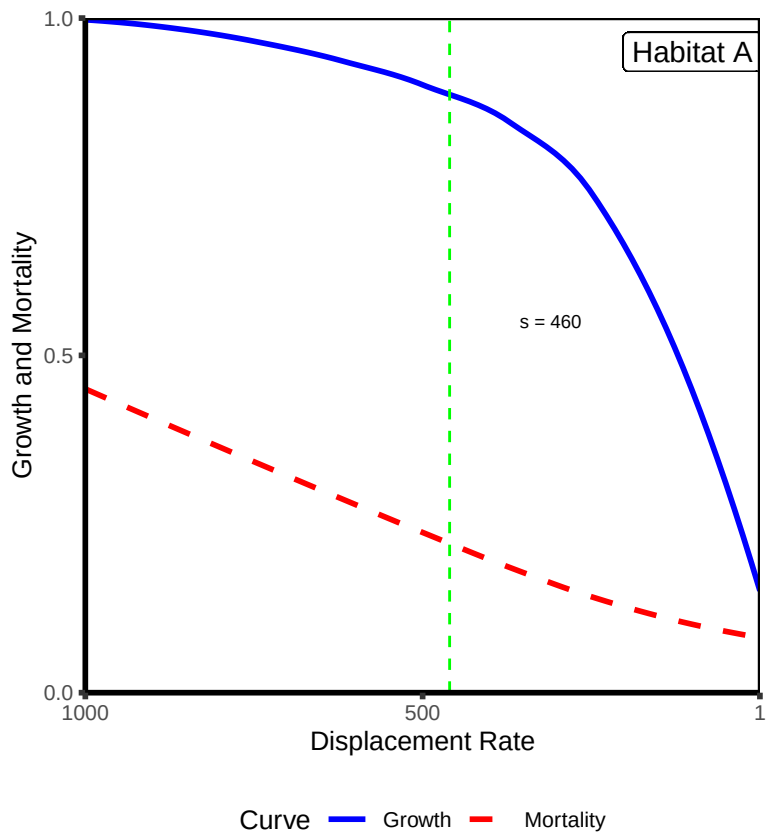
```
## The growth-trait curve R is prey resource density s is
## prey body velocity h is handling time

growth.curve <- function(s, h, R) {
```

```
    (s * R)/(1 + s * h * R)
}
```

Deriving Prey Speed in Habitats A and B

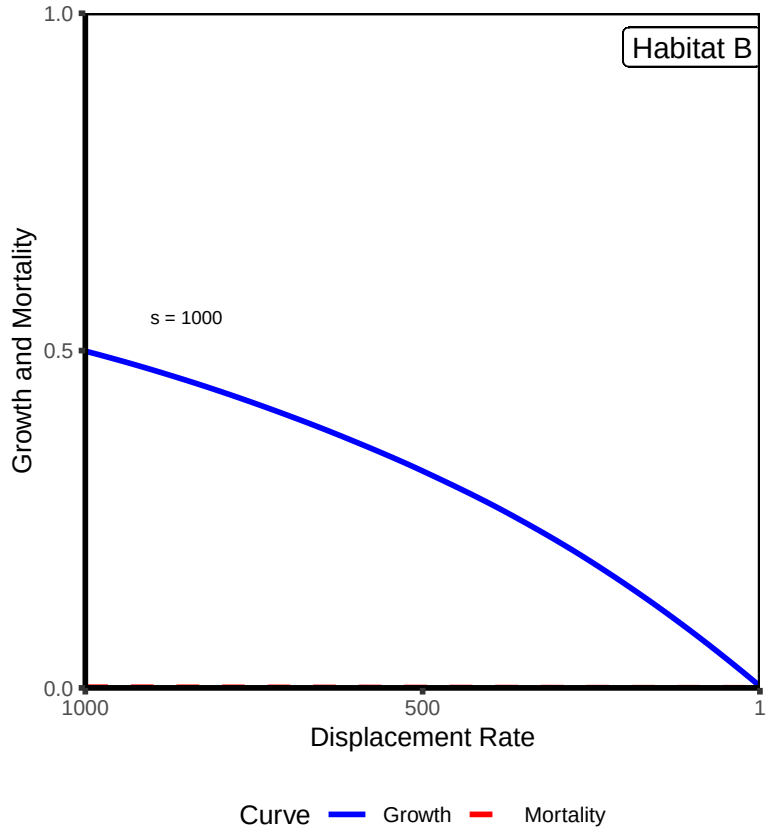
We will begin by determining the rate at which the prey should forage inside and outside of the refuge. We'll start by generating the curves for the habitat outside the refuge — we'll call it Habitat A. In Habitat A, predators are present at a moderate density ($k = 0.45$) but resources are plentiful ($R = 1$). We will assume prey handling time is constant, regardless of habitat choice ($h = 8$), and predator foraging rate will also be constant between habitats ($v = 0.2$).

This results in the following trait curves:

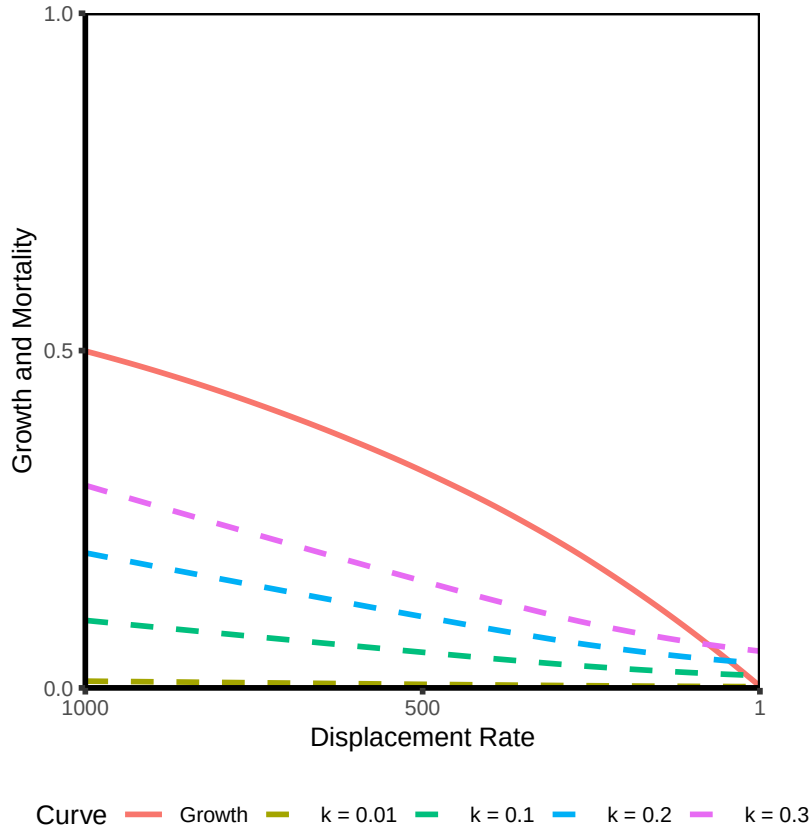


Thus, in Habitat A, fitness (1 - NCE - CE) is optimized at a displacement rate of ~460 ($s = 0.46$).

In Habitat B, the refuge, predators are very rare ($k_B = 0.001$) but resources are far less plentiful ($R_B = 0.05$). This results in the following trait curves:



In this set of curves, the optimal prey body velocity is at its maximum, in part because the mortality curve is very low and shallow. Having played around with the parameter space some, this seems to generally be the case for sufficiently low values of k_B, as shown below.



Even for $k_B = 0.3$, which is somewhat close to Habitat A's $k_A = 0.45$, the prey's optimal body velocity is still close to the maximum ($s = 0.96$).

Optimizing Time Spent in Refuge

Now that we've predicted the optimal body velocity for prey in Habitats A and B, given their resource and predator densities differ, we can think about how prey might optimize the amount of time they spend in either habitat across a given period of time. To do so, we will set prey body velocity constant at the levels derived above ($s_A = 0.46$, $s_B = 1$) and consider what degree of refuge use (on the x-axis) will maximize fitness.

The growth curve in this case can be calculated by simply adding the amount of resources the prey can gather during its time in the refuge vs. outside of it.

```
## Derive the trait curves
refuge.df <- as.data.frame(mat.or.vec(25, 5))
colnames(refuge.df) <- c("proptime", "growth_curve", "mort_curve",
    "fitness", "NCE_CE")
refuge.df$proptime <- seq(0, 1, by = 1/24)

s_val_A <- 0.46
s_val_B <- 1

# Growth is additive, so this is simply (growth during time
# outside refuge + growth during time inside refuge)
growth_A_val <- growth.curve(s = s_val_A, R = R_val_A, h = 8) *
    cost_constant_val  # Growth outside refuge during an hour
growth_B_val <- growth.curve(s = s_val_B, R = R_val_B, h = 8) *
    cost_constant_val  # Growth inside refuge during an hour

# Growth curve Growth is additive, so this is simply
# (growth during time outside refuge + growth during time
# inside refuge)
refuge.df$growth_curve <- (1 - refuge.df$proptime) * growth_A_val +
    (refuge.df$proptime) * growth_B_val
```

For the mortality curve, because risk is multiplicative instead of additive, we will calculate it using a weighted geometric mean:

$$\bar{x} = exp(\frac{\sum_{i=1}^{n} w_i lnx_i}{\sum_{i=1}^{n} w_i})$$

Because we are using a proportion to describe the amount of time spent in and out of the refuge habitat, $\sum_{i=1}^{n} w_i$ simplifies to 1, so the equation becomes:

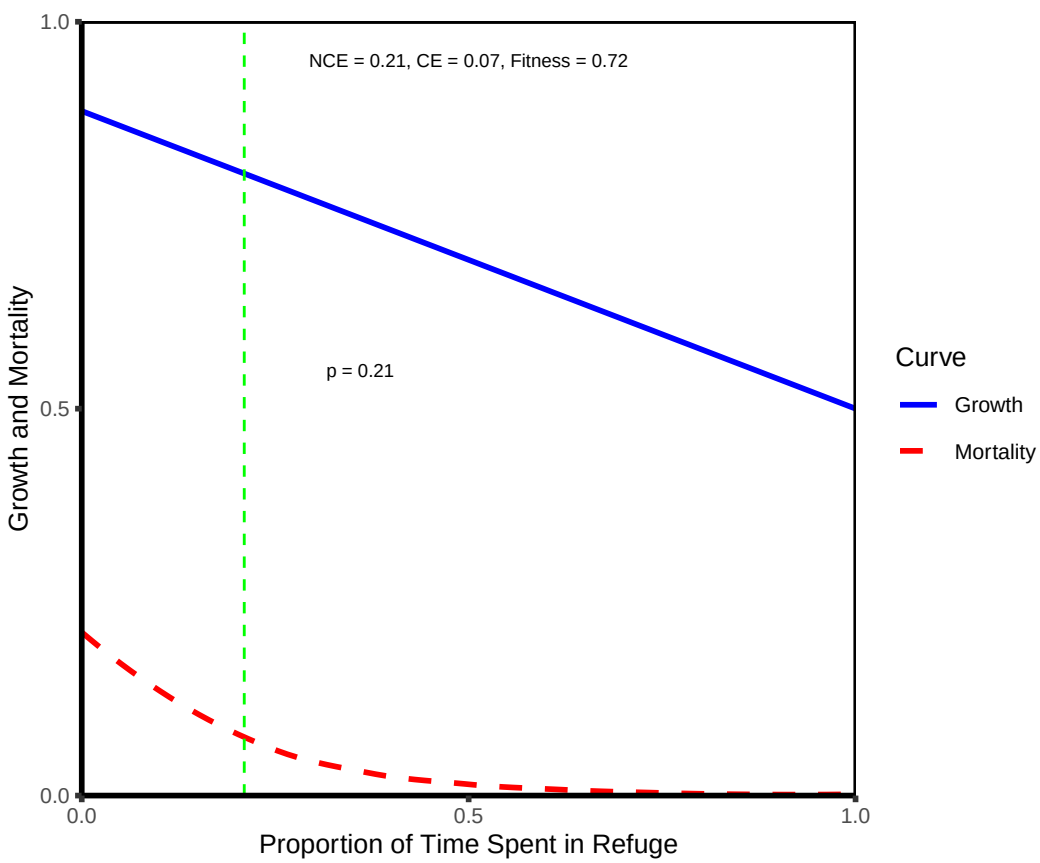
$$\bar{x} = exp(\sum_{i=1}^{n} w_i lnx_i)$$

Note that because this formula requires taking the natural log of the risk function, we cannot set predator density in the refuge (k_{refuge}) to zero without running into some trouble. So, I have instead set predator density to 0.001.

```
# Mortality curve
mort_val_A <- pred_constant_val * pred.curve(k = k_val_A, v = v_val,
    s = s_val_A)  # Mortality outside the refuge during an hour
mort_val_B <- pred_constant_val * pred.curve(k = k_val_B, v = v_val,
```

```
    s = s_val_B)  # Mortality inside the refuge during an hour

# Weighted mean of risk during time spent outside refuge +
# risk during time spent inside refuge
refuge.df$mort_curve <- exp((1 - refuge.df$proptime) * log(mort_val_A) +
    (refuge.df$proptime) * log(mort_val_B))

# Calculate fitness and relative strength of NCEs and CEs
refuge.df$fitness <- 1 - (1 - refuge.df$growth_curve) - refuge.df$mort_curve
refuge.df$NCE_CE <- refuge.df$growth_curve/refuge.df$mort_curve

# Some other important metrics
refuge_optim <- round(refuge.df$proptime[which(refuge.df$fitness ==
    max(refuge.df$fitness))], digits = 2)  # The level of refuge use (along x-axis) that fitness is opt
refuge_optim_fitness <- round(refuge.df$fitness[which(refuge.df$fitness ==
    max(refuge.df$fitness))], digits = 2)  # The fitness value at the optimum
refuge_optim_CE <- round(refuge.df$mort_curve[which(refuge.df$fitness ==
    max(refuge.df$fitness))], digits = 2)  # The strength of the CE at the optimum
refuge_optim_NCE <- round(1 - refuge_optim_CE - refuge_optim_fitness,
    digits = 2)  # The strength of the NCE at the optimum
```

This results in the following trait curves and optimal strategy.



In the above figure, the growth and mortality rates are shown as a function of the proportion of time that prey spend in the refuge. The blue line is the growth-trait curve, while the red dashed line is the mortality-trait curve. The optimal level of trait expression is noted with the vertical green dashed line.

The growth-trait curve is linear, because as time spent in the refuge increases, prey have proportionally less time to forage. On the other hand, the mortality-trait curve is saturating, because there are diminishing returns on how much prey can reduce their exposure to predators.

Incorporating Predator Foraging Mode

Looping back around to Os's work on the topic, a large part of his predictions had to do with what decisions a prey should make regarding foraging effort and refuge use given different predator foraging modes. In the above examples, I used $v = 0.2$, which is the level used in the grant proposal to represent a "sit-and-wait" style predator. Let's revisit his tables.

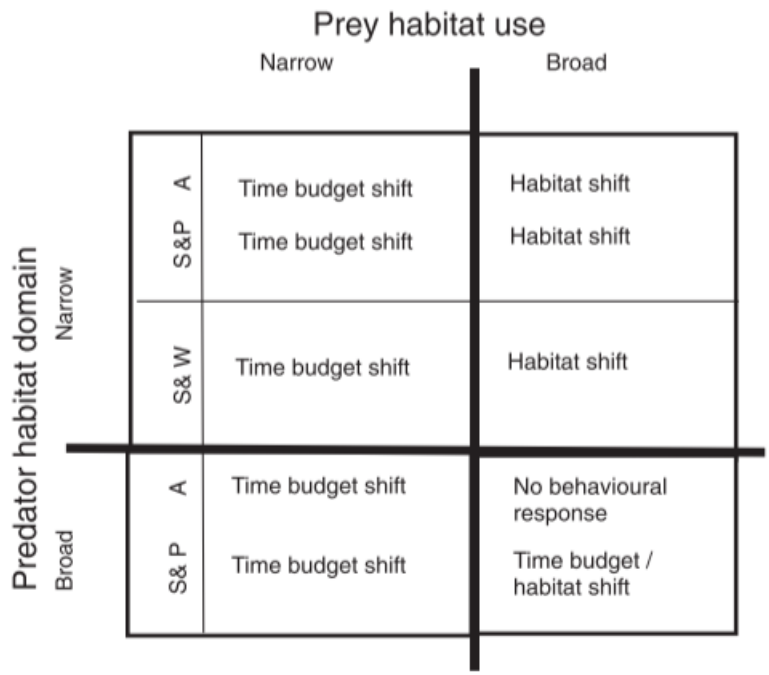


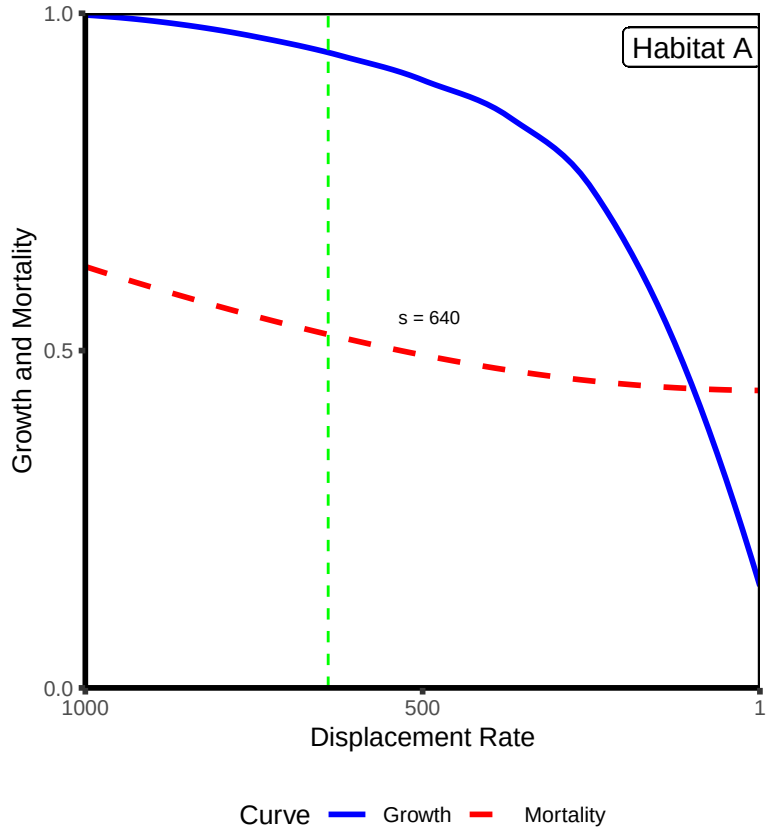
Figure 2 Synthesis of prey antipredator responses in relation to prey and predator habitat domain and predator hunting mode. S & P, A and S & W are sit-and-pursue, active and sit-and-wait hunting strategies, respectively. Species with a narrow habitat domain select only part of the entire available habitat. Species with a broad habitat domain use the entire range of habitat.

Figure 3: Fig. 2 from Schmitz et al. 2004

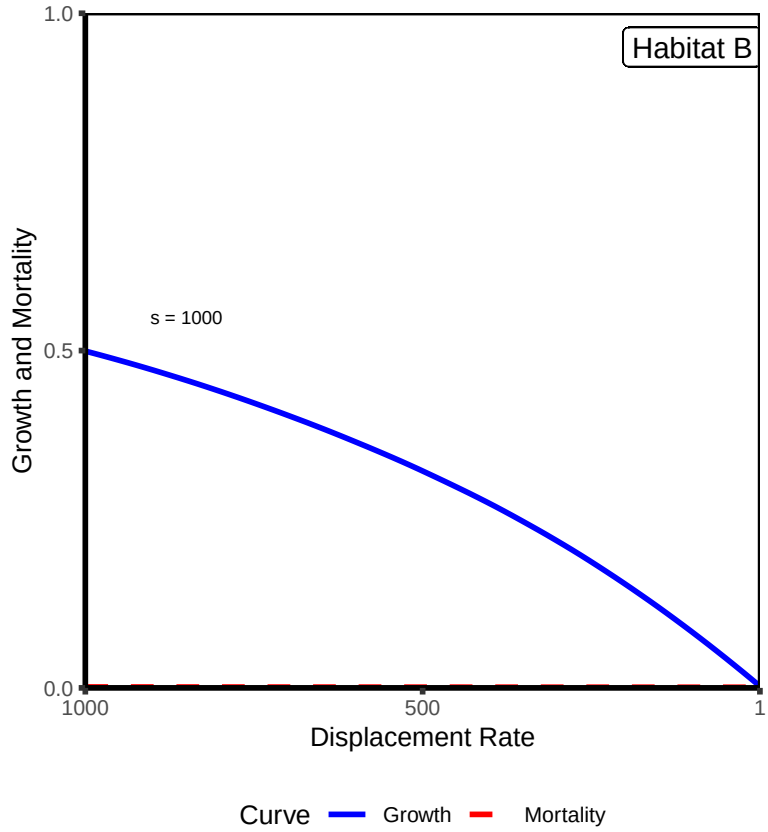
Os predicts that when the prey's habitat domain is larger than the predators (i.e., a refuge exists, the upper right quadrant), the prey should predominantly shift their habitat use to avoid predators regardless of their foraging mode. Here, we show that they should actually do a little bit of both depending on the relative risk and resource density present in either habitat.

But what if prey are faced with a more active predator? Does the optimal strategy change, or does our model still agree with Os's? Let's repeat the above steps with a predator with a much higher foraging rate ($v = 1$), as in the grant proposal, and see if it changes our predictions.

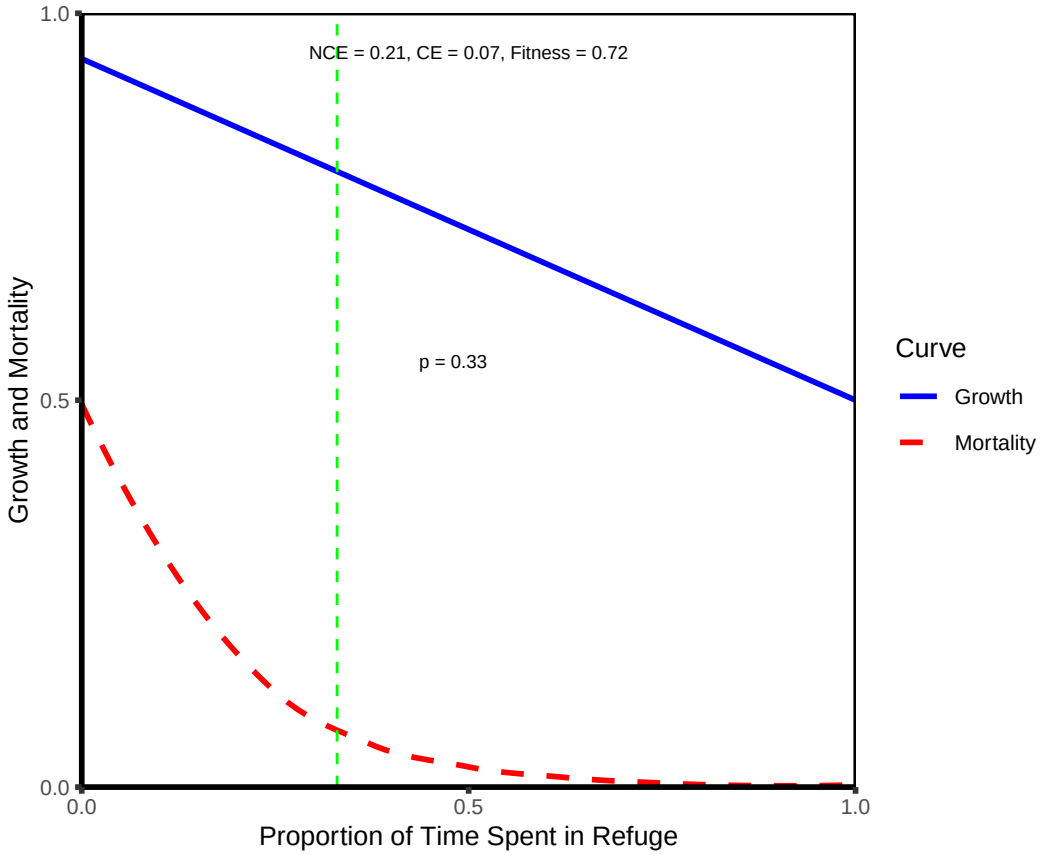
Outside the refuge (Habitat A), the prey's optimal body velocity changes as follows:



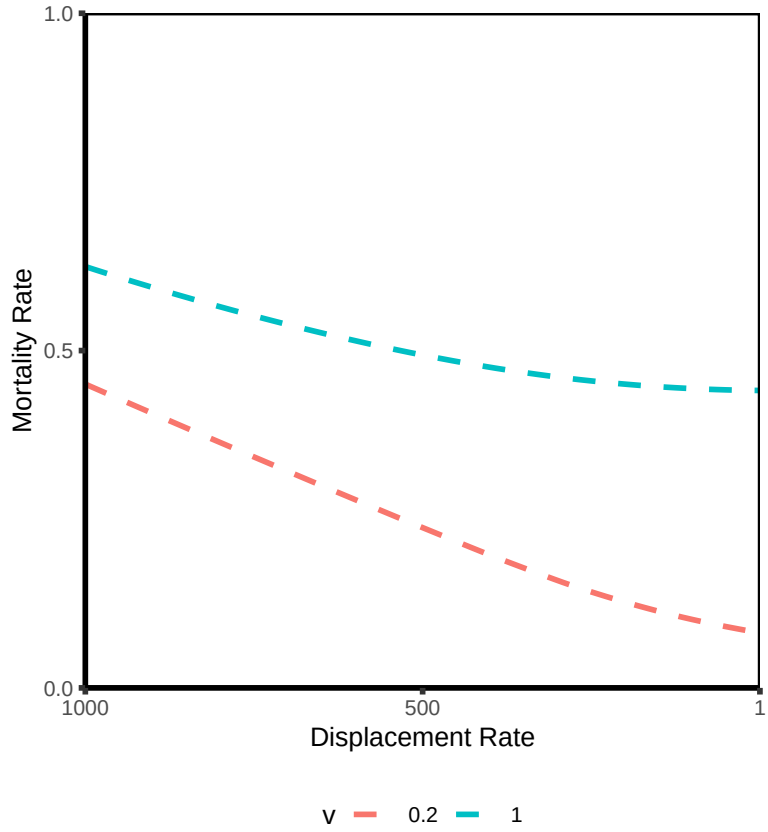
The optimum is higher, in line with what we expect from both Os's predictions and our own IREF predictions. Meanwhile, inside the refuge, the prey's optimal body velocity predator density is too low to shift the optimal prey velocity away from full speed.



And the optimal proportion of time spent in the refuge, given these prey body velocity values, is…



The IREF predicts that the prey should spend more time in the refuge faced with the active predator vs. the sit-and-wait predator ($p_{opt} = 0.33$ and $p_{opt} = 0.21$, respectively). This seems a little counter-intuitive at first, and may need some picking apart to explain. I think it relates in part to the fact that in this parameter space, a predator with a higher foraging rate is inherently more lethal. Consider the following mortality curves:



The mortality curve for predator with the higher body velocity is not only shallower, but it is shifted up along the y-axis, i.e., that predator is simply more lethal. So, due to this difference in lethality, the optimal solution is to spend more time "hiding out" in the refuge away from the predator with the higher search rate.

It may also line up with Os's predictions in his Fig. 3 above, where he suggests that a sit-and-pursue predator might have stronger DMIIs relative to TMIIs than an active predator (upper right quadrant). Note, however, that the calculated values for NCE and CE strength, as well as fitness, are basically the same between the two predator foraging modes.